



17 September 2026

DUNE Interface Control Document: Slow Controls, Photon Detector System and Detector Protection System

Summary

This document defines DAPHNE control, configuration and monitoring (CCM) between PDS, DAQ and Slow Controls over each board's shared 1 Gb Ethernet connection. DAQ CCM controls run variables and timing-endpoint settings through the configuration component integrated into the external OPC UA server. SC can use that path to recover timing and publish board readiness. Hermes remains a separate service on the same link.

Configuration uses request/response; monitoring is published. DAQ opmon receives firmware counters, timing state and applied readout configuration from `daphne-server` through OPC UA. SC monitors voltages, temperatures, fans, service status and versions. The inventory comprises 42 FD-VD boards and 150 FD-HD boards. DAQ data readout and optical timing transport are outside this scope.

DUNE Doc:	SC-PDS-DPS ICD	
EDMS Id:	To be assigned	
Version:	0.2 — control, configuration and monitoring	
Last modified:	17 September 2026	
Prepared by:	Checked by:	Approved by:
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1 Physical interface and service placement

Each DAPHNE provides one dedicated 1 GbE SFP connection for CCM, as specified in the DAQ-PDS ICD [R1, R2]. All board-facing services below share this link:

DAPHNE Ethernet interface $\longleftrightarrow$ switch $\longleftrightarrow$ physical server.

PDS supplies the fibres to its mini-rack patch panels; continuation and active switching are coordinated with DAQ and Technical Coordination [R1].

1.1 Detector inventory

Detector	PDS channels	Channels/board	DAPHNE boards
FD-VD	1,344	32	42
FD-HD	6,000	40	150

These are the detector design allocations [R2], with one CCM connection per board. FD-VD includes membrane and cathode modules; both use this CCM interface.

1.2 Service locations

Location	Services
DAPHNE board	daphne-server and ZeroMQ endpoints; local timing-endpoint access; Hermes IPBus/UDP configuration service.
External physical server	OPC UA server containing the DAQ configuration component, Protobuf/ZeroMQ client and monitoring subscriber.
SC server infrastructure	Ignition, acting as an OPC UA client of the external server.
DAQ server infrastructure	DAQ CCM client/opmon and the separate Hermes client.

Services may share a physical host. The installation record shall name their hosts, board/switch/server ports, SFPs and patch routes. OPC UA terminates on the external server. Board services, configuration and monitoring share the available GbE bandwidth.

2 Control, configuration and monitoring protocols

2.1 Service architecture

The OPC UA server hosts DAQ configuration and forwards publications from `daphne-server`: equipment monitoring to Ignition; firmware counters, timing state and applied readout configuration to DAQ opmon. Hermes receives configuration over UDP/IPBus.

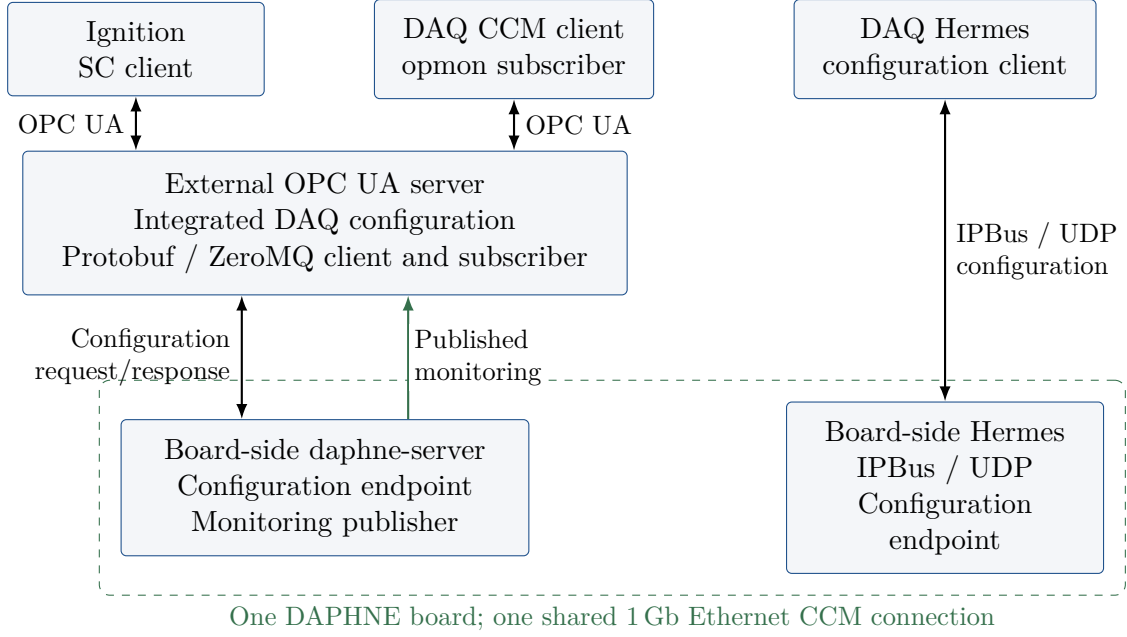


Figure 1: CCM configuration and published monitoring.

2.2 Configuration exchanges

DAQ CCM controls run variables and timing-endpoint settings through OPC UA. The server exchanges Protobuf messages over ZeroMQ and TCP using a DEALER client and the board's ROUTER endpoint [R3, R4]. The reference port is TCP 40001 [R5]. Correlated responses shall distinguish acceptance, completion and failure.

Hermes retains its independent IPBus/UDP configuration endpoint, reference UDP 50001 [R6], outside the OPC UA server. Both services share the board connection and gateway lifecycle.

2.3 Published monitoring

Monitoring shall be published periodically or on change, without a client request for each update. Only configuration and control operations, including reset/recovery commands,

use application request/response. Local hardware sampling is performed by the producing service.

The proposed publication transport is Protobuf over ZeroMQ PUB/SUB over TCP, using endpoints separate from configuration. The OPC UA server subscribes to **daphne-server** publications and distributes them through OPC UA subscriptions. Opmon monitoring follows the path **daphne-server** → OPC UA server → DAQ opmon.

Consumer	Monitoring responsibility
DAQ opmon	Firmware counters, timing state and applied data-readout configuration, published by daphne-server through OPC UA.
SC / Ignition	Voltages, temperatures, fans, service status and software/firmware versions.
SC and DAQ	Timing-endpoint condition, reset/recovery progress and board readiness.

Opmon shall not acquire voltage monitoring. SC receives equipment monitoring through the OPC UA server without an opmon dependency. Publications shall carry source identity, time, sequence and validity. Subscribers shall detect stale or missing updates and obtain a current snapshot after reconnecting.

2.4 Timing-endpoint control and recovery over GbE

Timing-endpoint configuration follows the DAQ configuration path:

DAQ $\longleftrightarrow$ OPC UA server $\longleftrightarrow$ DAPHNE service $\longleftrightarrow$ local timing endpoint.

The same interface shall expose reset and recovery to authorized SC clients. Timing monitoring shall be published over the board's GbE connection and made available to both SC and DAQ.

A reset request shall identify its target and receive an execution response. The board shall publish reset assertion, release and the resulting endpoint state so that both clients can acknowledge the same recovery operation. Recovery access shall remain available while timing is unavailable. Concurrent DAQ and SC commands shall be serialized.

SC shall determine and publish *board ready for data taking* from the agreed readiness conditions, including completed reset, restored clocks, endpoint readiness and valid synchronization [R4]. Reset release alone is insufficient. Missing or stale required publications shall withhold readiness. DAQ consumes the published readiness before resuming data taking.

2.5 Interface release and verification

Each release shall publish online the compatible server/client versions, Protobuf schemas, OPC UA model, Hermes interface and endpoint/port allocation. The configuration and

publication contracts shall specify authentication, timeouts, queue limits, update cadence and reconnect behavior. Publication schemas and ports remain to be assigned.

Verification shall cover concurrent configuration and publication on the shared link, the HD and VD installation scales, subscriber loss/reconnection, and SC timing recovery with a board-generated completion acknowledgement. The DAQ-PDS and DAQ-SC/DPS interface documents shall use this same CCM contract.

References

- R1** DAQ-PDS ICD, 2621471: CCM physical connection.
- R2** DAQ-PDS proposed v8, 6c0b245: document format, HD/VD inventory and configuration architecture.
- R3** daphnmodules, edf75a0: configuration schema and DEALER client.
- R4** daphne-os/daphne-server, e31bdfb: ROUTER configuration endpoint and endpoint recovery prerequisites.
- R5** daphne-os/petalinux, e31bdfb: board service endpoints and lifecycle.
- R6** hermesmodules, 887d5fc: independent IPBus/UDP configuration service.

The cited implementations supply the configuration baseline. The publication and SC recovery requirements specify the revised CCM architecture.